

## 6. Voice Input/Output (Bhashini / Web Speech)

---

**Target Goal:** Enable farmers to speak their crop questions and hear answers spoken aloud in their native tongue without typing.

### WHY VOICE IS ESSENTIAL FOR SMALL FARMERS

Many smallholder farmers have limited formal literacy or have wet, muddy hands while inspecting crops in the field. Voice interaction makes CropCare AI as natural as talking to an agricultural expert standing beside them.

### Method A: Zero-Cost Native Web Speech API (Works on all Android Phones)

---

No backend, server, or API key needed. Runs directly on device:

```
// src/hooks/useVoiceAssistant.ts
import { useState, useEffect } from 'react';

export function useVoiceAssistant(languageCode = 'hi-IN') {
  const [isListening, setIsListening] = useState(false);
  const [transcript, setTranscript] = useState('');

  // 1. Speech-to-Text (Farmer Speaking)
  const startListening = () => {
    const SpeechRecognition = (window as any).SpeechRecognition || (window as any).webkitSpeechRecognition;
    if (!SpeechRecognition) {
      alert('Speech recognition is not supported in this browser.');
```

```
      return;
    }
  }

  const recognition = new SpeechRecognition();
  recognition.lang = languageCode; // e.g. 'hi-IN', 'mr-IN', 'te-IN', 'ta-IN'
  recognition.continuous = false;
  recognition.interimResults = false;

  recognition.onstart = () => setIsListening(true);
  recognition.onend = () => setIsListening(false);
  recognition.onresult = (event: any) => {
    const text = event.results[0][0].transcript;
    setTranscript(text);
  };

  recognition.start();
};

// 2. Text-to-Speech (AI Doctor Speaking Back)
const speakText = (text: string) => {
  if (!('speechSynthesis' in window)) return;
  window.speechSynthesis.cancel(); // Stop ongoing speech

  const utterance = new SpeechSynthesisUtterance(text);
  utterance.lang = languageCode;
  utterance.rate = 0.95; // Slightly slower for clear agricultural comprehension
  window.speechSynthesis.speak(utterance);
};

return { isListening, transcript, startListening, speakText };
}
```

## Method B: Digital India's Bhashini AI (Government Open Speech Engine)

For rural dialects, Government of India's **Bhashini AI** offers the highest accuracy across Indian languages:

```

# backend/bhashini_service.py
import requests

BHASHINI_ENDPOINT = "https://dhruva-api.bhashini.gov.in/services/inference/pipeline"
API_KEY = "YOUR_BHASHINI_API_KEY"

def translate_and_speak(text: str, source_lang: str = "en", target_lang: str = "hi"):
    headers = {
        "Authorization": API_KEY,
        "Content-Type": "application/json"
    }
    payload = {
        "pipelineTasks": [
            {
                "taskType": "translation",
                "config": {
                    "language": { "sourceLanguage": source_lang, "targetLanguage": target_lang }
                }
            },
            {
                "taskType": "tts",
                "config": {
                    "language": { "sourceLanguage": target_lang },
                    "gender": "female"
                }
            }
        ],
        "inputData": { "input": [{ "source": text }] }
    }
    response = requests.post(BHASHINI_ENDPOINT, json=payload, headers=headers)
    return response.json()

```